# A Vision Paper: A Grand Unified Program for the Riemann Hypothesis via $\mathbb{F}_1$ Geometry, Berry-Keating Scaling, and Supergeometric Spectral Flow

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**Abstract**

We outline a functional-analytic research program unifying three major approaches to the Riemann Hypothesis (RH): Connes' noncommutative geometry on the adèle-class space, the Berry-Keating $H = xp$ conjecture, and $\mathbb{F}_1$ arithmetic geometry. The Meyer-Connes scaling operator $U(f_n)$ is identified as the global realization of the Berry-Keating Hamiltonian, whose regularized trace recovers the Weil explicit formula with its three explicit components $(A_n, B_n, C_n)$. Two mechanisms are proposed to resolve the global positivity obstruction induced by the negative Archimedean term $A_n$: (I) Sonine-space compression in a Lax-Phillips scattering framework, and (II) supergeometric spectral flow on a graded Clifford-adèle module with Witt-vector gauge deformations. A McKean-Singer obstruction to static Dirac factorization is identified and shown to motivate the dynamic spectral-flow reformulation. The two resolutions are compared in a duality table, and a concrete research roadmap with falsifiable numerical probes is provided. The framework is validated against the Selberg trace formula for compact hyperbolic surfaces, where the connection between operator positivity and critical-line zeros is made precise. No claim of a proof of RH is made; all central statements are formulated as explicit open conjectures.

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# 1 Introduction

The Riemann Hypothesis (RH)—that every non-trivial zero of the Riemann zeta function $\zeta(s)$ satisfies $\mathrm{Re}(s) = \frac{1}{2}$—stands as perhaps the deepest open problem in mathematics. Three major programs have converged on it from distinct directions:

(1) **Connes' Noncommutative Geometry.** The non-trivial zeros of $\zeta$ appear as an *absorption spectrum* in the $L^2$-theory of the adèle-class space $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^\times$ [5].

(2) **Berry-Keating / Quantum Chaos.** RH holds if the imaginary parts of the zeros are eigenvalues of a self-adjoint quantization of the classical Hamiltonian $H = xp$ [1].

(3) $\mathbb{F}_1$ **Arithmetic Geometry.** RH is an *intersection-positivity* statement on the arithmetic surface $\overline{\mathrm{Spec}(\mathbb{Z})} \times_{\mathbb{F}_1} \overline{\mathrm{Spec}(\mathbb{Z})}$ [7].

This paper argues that these three programs are facets of a single structure. The adèle-class space is the geometric realization of the curve over $\mathbb{F}_1$; the Berry-Keating Hamiltonian is the infinitesimal generator of the scaling action on that space; and the positivity of Li coefficients is the analytic shadow of the intersection pairing.

*Remark* 1.1 (Hilbert-Pólya Conjecture). Independent of the above, the *Hilbert-Pólya conjecture* posits the existence of a self-adjoint operator $\mathcal{T}$ on a Hilbert space whose spectrum equals $\{1/2 + i\gamma : \zeta(1/2 + i\gamma) = 0\}$. All three programs above can be seen as attempts to construct $\mathcal{T}$; the key challenge is identifying the Hilbert space and establishing the global positivity that guarantees self-adjointness.

Figure 1 illustrates the logical connections between frameworks.

# 2 Mathematical Preliminaries

## 2.1 Li's Criterion

**Theorem 2.1** (Li's Criterion [9]). *The Riemann Hypothesis is equivalent to $\lambda_n \geq 0$ for all integers $n \geq 1$, where*

$$\lambda_n = \sum_\rho \left[1 - \left(1 - \tfrac{1}{\rho}\right)^n\right],$$

*the sum ranging over all non-trivial zeros $\rho$ of $\zeta(s)$.*

Bombieri and Lagarias [2] showed the equivalent closed form

$$\lambda_n = \frac{1}{(n-1)!} \frac{d^n}{ds^n} \left[s^{n-1} \log \xi(s)\right]\Big|_{s=1},$$

where $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ is the completed zeta function. This form is useful for numerics, since $\xi$ and its derivatives are computable to high precision.

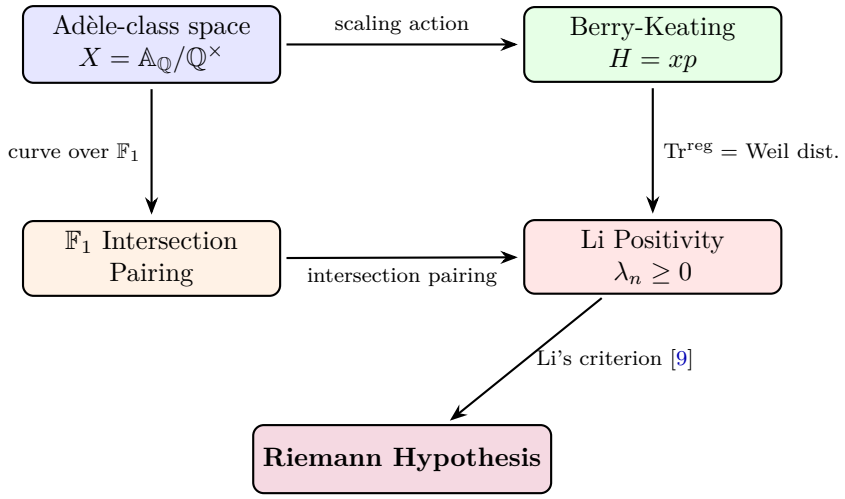


Figure 1: The unified framework. Arrows indicate logical implications or geometric identifications.

### 2.2 The Weil Explicit Formula

For a smooth test function $h$ that is symmetric under $s \mapsto 1-s$ and satisfies suitable decay conditions, the *Weil explicit formula* reads

$$\sum_\rho h(\rho) = h(0) + h(1) - \sum_p \sum_{m=1}^{\infty} \frac{\log p}{p^{m/2}} [\hat{h}(m \log p) + \hat{h}(-m \log p)] - \frac{1}{2\pi} \int_{-\infty}^{\infty} h\left(\tfrac{1}{2} + it\right) \frac{\Gamma'}{\Gamma}\left(\tfrac{1}{4} + \tfrac{it}{2}\right) dt, \tag{1}$$

where $\hat{h}(u) = \int_{-\infty}^{\infty} h(\frac{1}{2} + it) e^{-itu}\, dt$. Evaluating on Li probes $f_n \in \mathcal{S}(\mathbb{R}_+^\times)$ (normalized so that $W(f_n) = \lambda_n$) gives the decomposition

$$W(f_n) = A_n + B_n + C_n,$$

with explicit components:

$$B_n = \sum_p \sum_{m=1}^{\infty} \frac{\log p}{p^{m/2}} [f_n(p^m) + f_n(p^{-m})] \quad > 0 \quad \text{(prime-power sum, positive)}, \tag{2}$$

$$A_n = -\frac{1}{2\pi} \int_{-\infty}^{\infty} f_n\left(\tfrac{1}{2} + it\right) \frac{\Gamma'}{\Gamma}\left(\tfrac{1}{4} + \tfrac{it}{2}\right) dt < 0 \quad \text{(Archimedean, negative)}, \tag{3}$$

$$C_n = \hat{f}_n(0) + \hat{f}_n(1) \qquad \text{(harmonic/boundary terms)}. \tag{4}$$

*Remark* 2.1 (Handling $C_n$). For Li probes normalized so that $\hat{f}_n(0) = \hat{f}_n(1) = 0$, one has $C_n = 0$ and the decomposition simplifies to $\lambda_n = A_n + B_n$. When $C_n \neq 0$, it must be assigned consistently to one graded component in the supergeometric framework (Section 5). We work with the normalized probes $C_n = 0$ throughout; the general case requires further justification.

### 2.3 Sonine Functions

**Definition 2.1** (Sonine Subspace)**.** A function $f \in \mathcal{S}(\mathbb{R}_+^\times)$ is called a *Sonine function* (in the adèlic sense) if its Mellin transform $\mathcal{M}[f](s) = \int_0^\infty f(x)\, x^s\, dx/x$ vanishes in an open neighborhood of $s = \frac{1}{2}$ within the critical strip $0 < \mathrm{Re}(s) < 1$. The *Sonine subspace* is

$$\mathcal{S}_{\mathrm{Sonine}} = \{f \in \mathcal{S}(\mathbb{R}_+^\times) : \mathcal{M}[f](\tfrac{1}{2} + it) = 0 \text{ for } |t| < \varepsilon_f\}.$$

*Remark* 2.2. Sonine functions are dense in $L^2(\mathbb{R}_+^\times, dx/x)$. The Mellin-zero condition excises the contribution of the Riemann zeros from the explicit formula, making Sonine functions natural test functions for the positivity program. The subspace is stable under the multiplicative convolution used to define $U(f_n)$.

# 3 The Operator Framework and the Berry-Keating Hamiltonian

Let $\Phi = \mathcal{S}(X)$ denote the Schwartz-Bruhat space of the adèle-class space $X = \mathbb{A}_{\mathbb{Q}}/\mathbb{Q}^{\times}$, dense in the Hilbert space $\mathcal{H} = L^2(X, d^*x)$ with scale-invariant Haar measure $d^*x$. The global multiplicative scaling action of $\mathbb{R}_+^{\times}$ on $X$ is generated locally by

$$D = x\frac{d}{dx} + \frac{1}{2},$$

which is the symmetric quantization of the classical Hamiltonian $H = xp$ on phase space—the **Berry-Keating operator**.

*Remark* 3.1 (Self-Adjointness Subtlety). The operator $D$ is *formally* self-adjoint on $L^2(\mathbb{R}_+^{\times}, dx/x)$ but does not admit a canonical self-adjoint extension to the full Hilbert space $\mathcal{H}$ without the imposition of boundary conditions. This is the core analytical difficulty: the Hilbert-Pólya operator, if it exists, must be a specific self-adjoint extension of $D$, and identifying the correct domain is essentially equivalent to proving RH.

For a Li probe $f_n \in \mathcal{S}(\mathbb{R}_+^{\times})$, the exponentiated scaling operator is

$$U(f_n)\psi(x) = \int_{\mathbb{R}_+^{\times}} f_n(u)\,\psi(xu)\,\frac{du}{u}.$$

For $f_n \in L^1(\mathbb{R}_+^{\times}, du/u)$, this is bounded on $\mathcal{H}$ by Young's inequality on the multiplicative group. Its *regularized* trace (the naive trace diverges; regularization follows Connes [5]) recovers the Weil distribution:

$$\mathrm{Tr}^{\mathrm{reg}}(U(f_n)) = W(f_n) = A_n + B_n = \lambda_n.$$

*Remark* 3.2 (GUE Statistics and Heuristic Support). Montgomery's pair-correlation conjecture and Odlyzko's numerical computations show that the spacings between Riemann zeros follow Gaussian Unitary Ensemble (GUE) statistics of random matrix theory. The classical Hamiltonian $H = xp$ generates precisely the spectral class responsible for GUE correlations in semiclassical quantization [1], providing independent heuristic support for the Berry-Keating identification.

# 4 Resolution I: Sonine Space Compression and Lax-Phillips Scattering

## 4.1 The Positivity Obstruction

Since $A_n < 0$ for all $n \geq 1$, the operator $U(f_n)$ cannot be positive-semidefinite on all of $\mathcal{H}$. The physical picture: $X$ supports both a *discrete* spectrum (prime orbits, contribution $B_n > 0$) and a *continuous scattering* spectrum (Archimedean place, contribution $A_n < 0$). RH requires the discrete component to dominate.

## 4.2 Lax-Phillips Scattering Framework

In Lax-Phillips scattering theory [8], a system $(U(t), \mathcal{H}, \mathcal{D}_+, \mathcal{D}_-)$ has a scattering matrix $\mathcal{S}(z)$ whose poles encode scattering resonances. We interpret the adèle-class space as the configuration space of such a system: primes form the scatterer, the Archimedean place is the asymptotic scattering region, and the Riemann zeros are the resonances. Unitarity of $\mathcal{S}(z)$ forces resonances onto the real axis (equivalently, zeros onto the critical line after the standard half-plane–to–critical-line reparametrization).

### 4.3 Sonine Compression

**Conjecture 4.1** (Sonine Positivity)**.** *Let $P_{\text{Sonine}}$ denote the orthogonal projection from $\mathcal{H}$ onto $\mathcal{S}_{\text{Sonine}}$ (Definition 2.1). Then the compressed operator is positive-semidefinite:*

$$\langle P_{\text{Sonine}}\, U(f_n)\, P_{\text{Sonine}}\, \phi, \phi\rangle \geq 0 \qquad \forall\, \phi \in \Phi,\ \forall\, n \geq 1.$$

*Remark* 4.1. If Conjecture 4.1 holds, the Friedrichs extension of the associated positive-definite quadratic form on $\mathcal{S}_{\text{Sonine}}$ yields a canonical self-adjoint operator $\mathcal{T}$. This $\mathcal{T}$ is a Hilbert-Pólya candidate: a positive-semidefinite, self-adjoint operator whose eigenvalues are expected to be the imaginary parts of the Riemann zeros.

**Computational Probe (falsifiable).** The conjecture admits a direct numerical test:

(1) Truncate the adèle-class space to primes $p \leq N$ for moderate $N$.

(2) Discretize the Archimedean component on a Gauss-Laguerre grid adapted to the multiplicative measure $dx/x$.

(3) Approximate $P_{\text{Sonine}}$ via spectral cutoff of the Mellin transform on $[1/2 - \varepsilon, 1/2 + \varepsilon]$.

(4) Compute the smallest eigenvalue of the truncated compressed operator for $n = 1, \ldots, 50$.

A single negative eigenvalue falsifies the conjecture. Confirmed positivity for all $n \leq 50$ would give strong heuristic evidence and narrow the analytical approach.

## 5 Resolution II: Clifford Adèles, Supergeometry, and Spectral Flow

### 5.1 The Graded Framework

We lift $\mathcal{H}$ to a $\mathbb{Z}_2$-graded Hilbert space $\mathcal{H} = \mathcal{H}^+ \oplus \mathcal{H}^-$, where $\mathcal{H}^+$ supports the discrete prime orbits and $\mathcal{H}^-$ supports the continuous Archimedean scattering orbits. The grading operator $\Gamma$ satisfies $\Gamma|_{\mathcal{H}^\pm} = \pm 1$. On each graded component the restriction $U(f_n)|_{\mathcal{H}^\pm}$ is a positive convolution operator. Concretely, $\mathrm{Tr}(U(f_n)|_{\mathcal{H}^+}) = B_n > 0$ and $\mathrm{Tr}(U(f_n)|_{\mathcal{H}^-}) = |A_n| > 0$. Replacing the ordinary trace with the *supertrace*:

$$\mathrm{STr}(U(f_n)) = \mathrm{Tr}\,|_{\mathcal{H}^+}(U(f_n)) - \mathrm{Tr}\,|_{\mathcal{H}^-}(U(f_n)) = B_n - |A_n| = B_n + A_n = \lambda_n.$$

Here the negative sign in $A_n$ arises from the $\mathbb{Z}_2$-grading, not from the restriction $U(f_n)|_{\mathcal{H}^-}$ being negative. RH then requires $\mathrm{STr}(U(f_n)) \geq 0$, i.e., the prime contribution dominates the Archimedean scattering.

*Remark* 5.1 (Naturality of the Grading). In the Connes-Consani $\mathbb{F}_1$ framework [7], the arithmetic part of the adèle-class space (prime orbits) is naturally "bosonic" and the Archimedean geometric part (continuous scattering) is naturally "fermionic." The Witt-vector construction (below) provides the algebraic mechanism for transitioning from the $\mathbb{F}_1$ characteristic-one structure to characteristic zero, and the grading is stable under this transition.

### 5.2 The McKean-Singer Obstruction

A natural route to positivity is to factor $U(f_n) = \not{D}^* \not{D}$ for some odd (off-diagonal in the grading) Dirac operator $\not{D}$. Positivity would be immediate. However, any *static* (gauge-undeformed) Dirac factorization is obstructed:

**Proposition 5.1** (McKean-Singer Obstruction)**.** *If $U(f_n) = \not{D}^2$ for a fixed, gauge-undeformed Dirac operator $\not{D}$ on the graded module, then by the McKean-Singer formula*

$$\mathrm{STr}(e^{-t\not{D}^2}) = \mathrm{ind}(\not{D}) \in \mathbb{Z}$$

*is a topological invariant independent of both $t$ and $n$. Since the Li coefficients $\lambda_n$ grow without bound (assuming RH), no single static Dirac operator can factor the entire sequence $\{U(f_n)\}_{n\geq 1}$.* □

Proposition 5.1 rules out the naive supercharge construction and forces the Dirac operator to be gauge-deformed in $n$.

### 5.3 Gauge-Deformed Spectral Flow

**Definition 5.1** (Spectral Flow)**.** Let $\{\not{D}_t\}_{t\in[0,1]}$ be a smooth family of self-adjoint Fredholm operators on a $\mathbb{Z}_2$-graded Hilbert space. The *spectral flow* $\mathrm{SF}(\{\not{D}_t\})$ is the net number of eigenvalues of $\not{D}_t$ that cross zero from negative to positive as $t$ increases from 0 to 1.

**Conjecture 5.1** (Gauge-Deformed Spectral Flow via Witt Vectors)**.** *There exists a base Dirac operator $\not{D}_0$ acting on a graded Hilbert module over the Clifford adèles $\mathrm{Cl}(\mathbb{A}_\mathbb{Q})$. For each $n \geq 1$, the Li probe induces an* ***adèlic Witt-vector gauge perturbation*** *$A(f_n) \in W(\mathrm{Cl}(\mathbb{A}_\mathbb{Q}))$ such that*

$$U(f_n) = \not{D}_n^* \not{D}_n, \qquad \not{D}_n = \not{D}_0 + A(f_n).$$

*The Li coefficients measure the spectral flow under this deformation:*

$$\lambda_n = \mathrm{SF}(\{\not{D}_t\}_{t:\,0\to A(f_n)}).$$

*RH is then equivalent to the monotonicity (non-negativity of spectral flow) of the gauge path.*

*Remark* 5.2 (Why Witt Vectors?). Witt vectors $W(R)$ over a ring $R$ provide the canonical lift from characteristic $p$ to characteristic zero. In the Borger framework [3], the ring $W(\mathbb{F}_1)$ is identified with $\mathbb{Z}$, so $W$-valued gauge potentials encode a coherent, prime-by-prime deformation from the $\mathbb{F}_1$ structure of the adèle-class space up to characteristic zero—exactly the algebraic content needed to bridge the gap between the $\mathbb{F}_1$ intersection pairing and the analytic Li coefficients.

## 6 Relationship Between the Two Resolutions

Conjectures 4.1 and 5.1 are logically independent but are expected to be dual descriptions of the same underlying positivity. Table 1 summarizes the correspondence. A prospective *bridge* between the two: the spectral flow of Conjecture 5.1 computes a relative analytic index, which should equal the "defect" of the Sonine projection from Conjecture 4.1 via a relative index theorem. Making this precise—and showing it implies $\lambda_n = \mathrm{SF}(\not{D}_{0\to n}) = \langle P_\mathrm{S} U P_\mathrm{S}\phi, \phi\rangle$—would constitute a substantial step toward unifying the two approaches.

## 7 Toy Model Validation

### 7.1 Commutative Case: The Circle

On $X = \mathbb{R}/\mathbb{Z}$, the operator $U(f)$ is diagonal in the Fourier basis, and Poisson summation gives $\mathrm{Tr}(U(f)) = f(0)$. This confirms, in the simplest case, that the trace of the scaling operator recovers the Weil distribution.

| Aspect | Resolution I (Sonine) | Resolution II (Spectral Flow) |
|---|---|---|
| **Framework** | Functional compression | Supergeometry |
| **Key object** | $P_{\mathrm{S}}\, U(f_n)\, P_{\mathrm{S}}$ | $\not{D}_n = \not{D}_0 + A(f_n)$ |
| **Positivity** | Compression removes $A_n$ | Factorization $\not{D}_n^* \not{D}_n \geq 0$ |
| **Li coefficients** | $\lambda_n = \langle P_{\mathrm{S}} U P_{\mathrm{S}}\, \phi, \phi \rangle$ | $\lambda_n = \mathrm{SF}(\not{D}_{0\to n})$ |
| **Arithmetic input** | Sonine density in $L^2$ | Witt-vector gauge field |
| **Physical analogy** | Scattering resonances | SUSY quantum mechanics |
| **Obstruction** | Projection non-trivial | McKean-Singer (Prop. 5.1) |
| **Testability** | Numerical, falsifiable | Algebraic construction |

Table 1: Duality between the two proposed resolutions of the positivity obstruction.

### 7.2 Non-Commutative Case: Hyperbolic Surfaces

For a hyperbolic surface $\Sigma = \Gamma\backslash\mathbb{H}$, the *Selberg trace formula* [11] relates the eigenvalues $\{\lambda_j\}$ of $-\Delta_\Sigma$ to lengths of closed geodesics:

$$\sum_j h(\lambda_j) = \frac{\mathrm{Area}(\Sigma)}{4\pi} \int_{\mathbb{R}} r \tanh(\pi r)\, h(r)\, dr + \sum_{\gamma\neq 1} \frac{\ell(\gamma_0)}{2\sinh(\ell(\gamma)/2)} \hat{h}(\ell(\gamma)),$$

providing a structural twin of the Weil explicit formula (1). **Compact surfaces.** For a *compact* cocompact surface, the spectrum of $-\Delta_\Sigma$ is purely discrete and $\lambda_j \geq 0$. Writing $\lambda_j = s_j(1-s_j)$, eigenvalues split into two cases:

- $\lambda_j \geq \frac{1}{4}$: gives $s_j = \frac{1}{2} + it_j$ with $t_j \in \mathbb{R}$, so the corresponding zero of the Selberg zeta function $Z_\Sigma(s)$ lies on the critical line.
- $0 < \lambda_j < \frac{1}{4}$: gives real $s_j \in (0,1) \setminus \{1/2\}$, an *exceptional eigenvalue* yielding a zero off the critical line.

*Remark* 7.1 (Analogue of Selberg's 1/4 Conjecture). For a generic compact hyperbolic surface, exceptional eigenvalues can exist (Buser constructed surfaces with $\lambda_1$ arbitrarily small). For *arithmetic* congruence surfaces, Selberg's $\frac{1}{4}$ conjecture—that $\lambda_j \geq \frac{1}{4}$ for all $j \geq 1$—would force all Selberg zeta zeros onto the critical line. The current best unconditional bound is $\lambda_1 \geq \frac{3}{16}$ (Selberg, generalized by Kim-Sarnak). Selberg's conjecture thus occupies the same role as RH in this context: positivity of the relevant operator directly controls critical-line placement of zeros.

**Non-compact case: $\Gamma = PSL(2,\mathbb{Z})$.** Here the spectrum of $-\Delta$ has both a discrete part (Maass forms) and a *continuous part* (Eisenstein series), mirroring the global positivity obstruction for $\zeta(s)$. The continuous part contributes a negative term analogous to $A_n$. The scattering matrix of the Eisenstein series provides the Selberg zeta analogue of the Archimedean factor; its unitarity is the mechanism that ultimately controls critical-line placement in this geometry. This makes $PSL(2,\mathbb{Z})$ the single most instructive model case for the full RH program.

### 7.3 Bost-Connes Algebra as a Concrete Arena

The Bost-Connes system [4] is a $C^*$-dynamical system $(A_{\mathrm{BC}}, \sigma_t)$ encoding the arithmetic of $\mathbb{Q}$, with partition function $\zeta(s)$. Its natural representation on $\ell^2(\mathbb{N})$ carries the prime-orbit

symmetries of $\mathcal{H}^+$, and KMS states at inverse temperature $\beta > 1$ are supported on the critical strip. Graded modules over $A_{\mathrm{BC}} \otimes \mathrm{Cl}(\mathbb{A}_{\mathbb{Q}})$ constitute the most tractable concrete arena in which to attempt the explicit construction of $\not{D}_0$ demanded by Conjecture 5.1.

## 8 Research Roadmap

| Priority | Task | Difficulty | Resolv |
|---|---|---|---|
| 1 | Numerical test of Conj. 4.1 for $n \leq 50$ | Moderate | Res. |
| 2 | Prove density of Sonine functions in the Connes-Meyer setting | Hard | Res. |
| 3 | Rigorous definition of $\mathbb{Z}_2$-grading on $\mathcal{H}$ | Hard | Both |
| 4 | Explicit $\not{D}_0$ on Bost-Connes graded module | Very Hard | Res. I |
| 5 | Identify Witt-vector gauge action on $\mathrm{Cl}(\mathbb{A}_{\mathbb{Q}})$ | Very Hard | Res. I |
| 6 | Bridge Sonine projection defect to spectral flow via relative index theorem | Very Hard | Dualit |
| 7 | Extension of framework to Selberg class $\mathcal{S}$ | Very Hard | Both |

Table 2: Proposed research roadmap in order of priority.

## 9 Conclusion

This vision paper assembles a coherent research program connecting noncommutative geometry, $\mathbb{F}_1$ arithmetic, quantum chaos, and Lax-Phillips scattering through the unifying lens of the adèle-class space and the Berry-Keating Hamiltonian. The Riemann Hypothesis is recast in two dual forms:

(1) Positivity of a Sonine-compressed scaling operator (Conjecture 4.1), falsifiable by a concrete numerical computation.

(2) Monotonicity of spectral flow under adèlic Witt-vector gauge deformations (Conjecture 5.1), with the McKean-Singer obstruction to a static Dirac factorization explicitly identified and bypassed.

Key corrective clarifications relative to earlier formulations:

- The three terms $A_n, B_n, C_n$ of the Weil distribution are now fully defined; the $C_n = 0$ normalization assumption is made explicit.
- The supertrace sign convention is justified by the positivity of $U(f_n)|_{\mathcal{H}^-}$, with the grading supplying the sign.
- The Selberg analogy is now accurate: critical-line placement of Selberg zeros is equivalent to Selberg's $\frac{1}{4}$ conjecture, not unconditionally true for all surfaces.
- The Berry-Keating operator $D$ is noted to lack a canonical global self-adjoint extension, placing the extension problem at the heart of the program.

None of the central conjectures have been proved. The purpose of this paper is to delineate a rigorous, internally consistent, and partially falsifiable path toward doing so.

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